

27 Ans: A

$$eV_S = hf + \phi \rightarrow \phi = eV_S - hf$$

λ is halved $\rightarrow f$ is doubled.

$$eV_{S'} = h(2f) + \phi$$

$$eV_{S'} = h(2f) + (eV_S - hf)$$

$$eV_{S'} = eV_S + hf$$

λ is halved $\rightarrow f$ is doubled

but I is kept constant

so N/t is halved for this source. $I = (N/t) hf/A$

so total current $\sim$ total N/t due to both sources $= 1.5 N/t = 1.5 I_e$